EDGAR COLÓN BARRIENTOS

SENIOR FULL STACK SOFTWARE ENGINEER

Cloud-native architecture · Solutions Architect · Software development · DevOps · Process automation

PROFESSIONAL PORTFOLIO https://edgar-colon.mx

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PROFESSIONAL SUMMARY

Senior Full Stack Software Engineer specializing in cloud-native systems, with **experience since 2011** creating, modernizing, and improving web applications. I am passionate about turning highly demanding business requirements into intuitive, scalable, and robust products centered on **user experience (UX)** and built with excellence in every line of code. I design secure, highly available solutions while balancing architecture, business needs, software quality, performance, reliability, security, and cost optimization.

I am a self-directed learner, critical thinker, and **creative problem solver** with strong **analytical** and abstract reasoning skills. I am committed to **continuous learning and improvement**. My logical approach helps me adapt quickly to different programming languages, technologies, and paradigms, while **integrating effectively into cross-functional teams**.

I currently **elevate my engineering impact** through the **strategic use** of **AI** and **Spec-Driven Development (SDD)**, defining clear **specifications** to **guide AI coding agents in line with Clean Code principles**, **scalable architecture**, and **shift-left security**. I **combine these practices with the engineering judgment developed throughout my career** to **accelerate** the **delivery** of **secure**, **maintainable**, and **scalable applications**, addressing these priorities early in the SDLC while keeping **usability** and **UX** at the forefront.

My experience spans the complete software development life cycle (SDLC), from architecture through implementation, unit, integration, and regression **testing**, **deployment**, **operations**, and **maintenance**. I apply **Clean Code, Clean Architecture**, **SOLID principles**, and **design patterns to build maintainable**, **decoupled**, **scalable**, and **testable solutions**. I integrate frontend, backend, cloud services, and deployment automation, and I also develop workflows, process automations, and AI **agents with n8n, MCPs**.

I **incorporate cybersecurity principles** from the earliest design stages, applying a Security by Design approach and **OWASP Top 10 practices** to prevent common vulnerabilities in web applications and APIs.

I design software architectures based on the **pillars of cloud architecture**, as well as **microservices**, **serverless architectures**, **and event-driven architectures (EDA)**. I use Amazon S3, EventBridge, AWS Lambda with **Python** and **NodeJS**, and Amazon **SQS** to build scalable **microservices** prepared for high-demand workloads. I also manage container deployments on Amazon **ECS with AWS Fargate** and endpoint administration through Amazon **API Gateway**. For security, I implement strong authentication and authorization mechanisms with JWT using RS256, **custom authorizers**, and API keys to protect services and resources. I also use **bastion hosts or jump boxes** to control secure access to databases containing sensitive information.

On the frontend, I have advanced experience developing enterprise applications with **Vue (v2 - v3)**, **Angular (v4 to v22)**, **React (v16 - v19)**, responsive web design**, advanced CSS**, Tailwind CSS, and Bootstrap, as well as usability and user **experience (UX) principles**. I have developed real-time point-of-sale systems with **Socket.IO**, application-to-application integrations, and debit and credit card payment-processing solutions with **OpenPay**. I have also implemented transactional email capabilities with **SendGrid** and **Amazon SES (Amazon Simple Email Service)**.

I **implement DevOps and infrastructure-as-code (IaC) strategies** to provision architecture with **Terraform**, using a **multi-cloud approach**, and AWS CloudFormation across development, staging, and production environments. I use Bitbucket Pipelines, GitHub Actions, and Jenkins to automate **CI/CD** and software validation, while also incorporating **high availability and disaster recovery plans (DRP)** automated with **Terraform**.

I integrate **static analysis** and **code-quality tools** such as **SonarQube** into Bitbucket and Jenkins pipelines to establish quality gates, identify vulnerabilities and code smells, reduce technical debt, and improve software maintainability.

In addition to the services I use to implement microservices, serverless solutions, containers, **and event-driven architectures**, my AWS experience includes Amazon **S3**, Amazon **EC2**, **Elastic Beanstalk**, **Amazon RDS**, AWS **CloudFormation**, Amazon **SNS**, and Amazon **CloudWatch**. I administer these services through the **AWS Management Console, AWS CLI**, and **Terraform**, implementing asynchronous processing and solutions designed for high-demand scenarios.

I also have **experience creating cloud contact centers** with **Amazon Connect** and using **Contact Lens** to analyze conversations, perform sentiment analysis, and assess the customer experience after each call.

I have deployed **static websites using Amazon S3** or **GitHub Pages** as **hosting**, integrating them with domains managed through **GoDaddy** or Amazon **Route 53 and managing DNS through Cloudflare**. I configure name servers, DNS, and **Google Workspace** services while applying **security**, **availability**, threat **mitigation**, and **cyberattack protection practices through Cloudflare.**

I have experience designing, normalizing, and integrating **relational and NoSQL databases,** working with **PostgreSQL**, **MySQL**, **SQL Server**, and **MongoDB**. My background also includes deploying scalable cloud solutions with Amazon **DynamoDB** and Google **Cloud Datastore**, as well as operating highly available systems on **Amazon RDS and Amazon Aurora**.

On **Google Cloud Platform** (GCP), I have used services such as Google **Cloud Functions**, **Cloud Run functions**, **App Engine**, **Cloud Run**, **Cloud Build**, **Cloud Scheduler**, **Cloud Storage**, **Cloud SQL**, and **Cloud SQL Auth Proxy**. I also have experience administering and deploying resources through pipelines and the **Google Cloud CLI (gcloud).**

I complement my software engineering background with administration of **Linux/Unix servers, VPS instances, and virtual machines** on Amazon Web Services and Google Cloud Platform through secure **SSH connections**. In addition to ensuring infrastructure stability and high service availability, I implement **monitoring**, log **analysis**, **event logging**, and **observability** practices through **Amazon CloudWatch and custom dashboards**, allowing me to diagnose and resolve incidents efficiently in enterprise production environments.

TECHNICAL SKILLS

Backend

JavaScript, REST APIs, NodeJS with Express or NestJS; **Python** with **Flask**, **Django**, and **FastAPI**, **Java** with (**Spring**, **Spring Boot** RESTEasy), and PHP with Laravel using MVC. Development of **serverless microservices** with **AWS Lambda in Python 3** or Node.js, **Amazon API Gateway**, and Amazon **SQS** for processing queues.

Service integration: **SendGrid** for sending email through its API, Twilio for SMS delivery, and experience with the **Slack SDK and webhooks, AI-First Development**, **Spec-Driven Development (SDD)**, **Claude Code**, **SpecKit**

Frontend

HTML, CSS, **responsive design**, **Flexbox**, **CSS Grid Layout**, JavaScript, **Vue v2, v3**, **Angular** and **React**, I've used Styled Components, Bootstrap, Tailwind CSS, Material UI, and Element UI, **flutter**.

API and Web Service consumption, real-time applications, and **SEO (search engine optimization).**

Cloud and infrastructure - AWS

Creation of software architecture and flow diagrams, including the **design of communication between microservices**.

Use of **Terraform** and CloudFormation to manage infrastructure as code efficiently and at scale, including disaster recovery plans (DRP).

Automated deployments of Lambda functions, **Lambda Layers**, and services on **Amazon ECS** with **AWS Fargate**.

Administration of AWS IAM, RDS, **Aurora**, and EC2. I configure virtual private clouds (**VPCs**) from scratch, establishing subnets, security groups, and route tables to provide the foundational infrastructure and secure connectivity required to deploy cloud resources.

Databases

SQL Server 2008 and later versions, **MySQL**, and **PostgreSQL** on **AWS RDS and Amazon Aurora**.
Extensive experience in architecting and optimizing **high-volume databases**, ensuring **scalability** and peak **performance**.
NoSQL: **MongoDB**, Amazon **DynamoDB**, Google Cloud **Datastore**; **Redis**.

Architecture, methodologies, and additional skills

I **implement software architectures** and use **design patterns** and **principles** such **as Domain-Driven Design**, **hexagonal architecture**, **Clean Architecture**, **MVC**, **SOLID**, and **Clean Code**.

Experience successfully applying agile methodologies such as **Scrum**, **Agile**, and **Kanban** for efficient **project management**, using **Jira**, **Trello**, and **GitFlow**.

WebSockets, **GraphQL**, object-oriented programming, component-oriented programming, functional programming, and SQL.

Agentic AI & Methodologies: Spec-Driven Development (SDD), GitHub **SpecKit**, **Claude Code**, **Agentic Skills & Harnesses**.
Proficiency with Sketch, **Figma**, and Miro.

Languages

English - Intermediate: reading 90%, writing 80%, and speaking 70%.
I reached the intermediate level while studying at Harmon Hall.

PROFESSIONAL EXPERIENCE

Google Cloud Developer | BBVA (contract) · GoNet USA / Orion Innovation

Feb 2024 - Jul 2026

Hybrid (two days per week onsite) · Mexico City, Mexico

Tools and technologies: GCP Google Cloud Platform, Cloud SQL, Proxy de Cloud SQL, Looker Studio, BigQuery, Cloud Datastore, Java v8 to 21.

Member of BBVA's **Servicing Talent & Culture** engineering team.

- **Resolved critical compatibility issues during Java 11+ modernization** initiatives by upgrading or replacing deprecated and unsupported dependencies. This included migrating persistence layers from **Hibernate/JPA** to **MyBatis** and transitioning **Java EE APIs from the javax.* namespace to Jakarta EE's jakarta.*** namespace.
- Maintained and enhanced applications across multiple technology generations, including **Java 8, 11, and 17**; legacy **Python 2** and modern **Python 3** applications, both standalone and **Flask-based**; and **WordPress 4–6** platforms operating across legacy and modern **PHP environments spanning versions 4 through 8**.
- Developed new solutions using **Java 17 and 21, Python 3, Node.js, Angular 15+, Vue 3, and WordPress 6+**.
- Deployed and operated applications built with **Java 8 through Java 21** on **Google App Engine** within **Google Cloud Platform (GCP)**.
- Remediated vulnerabilities identified through **BBVA Red Team ethical-hacking assessments** by implementing **end-to-end encryption (E2EE)**, strengthening session validation, and enforcing **role-based access control (RBAC)** across both frontend and backend components. These measures improved sensitive-data protection and reduced the applications' attack surface.
- Rebuilt the **BBVA Insurance website** from the ground up using **Vue 3**, translating **Figma designs** into a fully functional, responsive, and user-centered web application.
- Modernized a complex legacy **email marketing platform** originally built with **AngularJS 1.6.9, Bower Components, Java 8, and SendGrid**. Migrated the backend to **Java 21**, adopted **RESTEasy** and **Swagger/OpenAPI**, and transitioned the codebase from `javax.*` to `jakarta.*`, improving security, maintainability, and long-term platform support.

- Implemented and maintained **CI/CD workflows** using **Bitbucket Pipelines, Jenkins, Google Cloud Build, SonarQube, and JFrog Artifactory**.
- Configured **SDKMAN** and **jEnv** to manage multiple local Java versions, from **Java 8 through Java 21**, enabling fast project-specific environment switching and parallel development across applications with different runtime requirements.

Lead Developer · Cloud Architect · DevOps | UBIQUIUS · Banpay.com

Apr 2023 - Jan 2024

On-site · Mexico City

Tools and technologies: SQL, **AWS Lambda**, **Amazon S3**, **Cloudflare**, Amazon **ECS**, **Python**, **FastAPI**, Django, Node.js, Vue, UML, **Sumsub**, **Terraform**, VPN, VPC, JWT RS256, Bitbucket, Git, Slack, Jira, Amazon EC2, SNS, SQS, RDS, **AWS Secrets Manager**, **software design principles** and **patterns**, and GitFlow.

- Helped onboard new team members by providing guidance and training on technologies such as **AWS Lambda**, API **Gateway**, and Amazon **ECS** / **Fargate**. I advised them as they became familiar with these tools and learned to use them effectively in their daily work.
- Managed users, groups, roles, and policies in **AWS Identity and Access Management (IAM)**.
- Built the **Banpay.com landing page** using Vue 3 and Pinia for state management. **Hosted** the site in an **Amazon S3 bucket configured for static website hosting and served it through Amazon CloudFront.** Used **Terraform** to manage the infrastructure and **Bitbucket Pipelines** to automate deployments across **development, QA, and production environments**. Initially managed the domain with Amazon Route 53, then moved DNS management to Cloudflare.
- Automated deployments of **Lambda functions** and **AWS ECS / Fargate** services across different environments. I implemented **AWS CLI** and managed infrastructure and code updates with **Terraform**, simplifying and improving the coordination of deployment operations.
- Implemented **JWT token creation** and **validation** using the **RS256** algorithm to strengthen API Gateway security. I configured a **custom authorizer** and implemented the sign-in process in Python. I also established a system that stored certificates in an Amazon S3 bucket and supported certificate rotation.
- Built backend services with **Python 3 and FastAPI**, **containerized with Docker**. Implemented **JWT-based authorization using the RS256 algorithm**, with API Gateway authorizers and FastAPI middleware to protect selected routes. **Automated deployments to Amazon ECS on Fargate through Bitbucket Pipelines**, including building Docker images, **publishing them to Amazon ECR**, and updating **ECS services**.
- Implemented static code analysis with **SonarQube** using a **Docker** container. I deployed the container to an **Amazon EC2** instance and connected it to a **subdomain managed through Cloudflare**.
- Implemented AWS services including **EventBridge**, SNS, SQS, and Lambda for selected cron jobs.
- In Cloudflare, I configured name servers for domains registered with **GoDaddy** and **Amazon Route 53**, managed DNS records, associated Amazon S3 bucket URLs and **AWS load balancers** with domains or subdomains, and established configurations to **mitigate DDoS attacks and traffic originating from TOR**.

Architecture and microservices delivered

Microservices architecture

Designed and developed a decoupled microservices architecture, primarily in **Python 3** with selected services in **Node.js**. Some microservices were invoked through **API Gateway**, while others were used **privately through a custom function** named lambdaInvoke

Identity Verification Service (KYC, KYB, KYT)

Developed a **serverless Python backend** to integrate with Sumsub's APIs for **KYC**, **KYB**, and **KYT identity verification**. Built the frontend with **Vue 3 and Pinia**. Used AWS Lambda and **Amazon API Gateway** for the backend, and implemented a **webhook** flow with Lambda, API Gateway, and Amazon **DynamoDB to process and store verification updates**. **Hosted** the frontend in an **Amazon S3 bucket configured for static website hosting, with DNS managed through Amazon Route 53**.

Serverless MFA Microservice

Designed and built a **serverless multi-factor authentication (MFA) service** in **Python** using the PyOTP library. It generated **QR codes** that users could scan with authenticator apps such as **Google Authenticator** or **Authy**. The architecture used AWS Lambda, Amazon API Gateway, Amazon DynamoDB, and Amazon S3. **To keep QR codes temporary** and **control** storage **costs**, stored them in S3, **provided time-limited access through pre-signed URLs**, and configured **S3 Lifecycle policies to delete them automatically**.

Email and SMS OTP Services - generation and validation

Built serverless services to **generate and validate one-time passwords (OTPs)** for authentication and MFA **via email or SMS**. The solution used AWS **Lambda**, Amazon **API Gateway**, Amazon **DynamoDB**, Amazon Route 53, Amazon **SNS**, Amazon **SES**, and **Amazon Pinpoint**.

Transactional Email Notification Microservice

Developed a **transactional email notification microservice** using **Amazon SQS** and **AWS Lambda** to process email requests asynchronously. A Lambda function consumed messages from the queue, selected the appropriate template based on the emailIdentifier logic, and sent the email through the **Twilio SendGrid API**. This **design decoupled email delivery** from the calling application and supported reliable processing of notification requests.

SMS and WhatsApp notifications

Designed and built a **notification microservice** using **AWS Lambda and Amazon SQS** to process **SMS** and **WhatsApp** notification requests asynchronously and deliver them through **Twilio's APIs**.

Custom database migrations

Designed and developed a custom database migration system in **Python** that could run from a **Lambda function** and manage database schema changes.

Used SSH keys and a bastion host to enable secure database connections from different locations.

Configuring the Lambda function inside a VPC, together with security groups, subnets, and other networking components, simplified migration execution and removed the need for manual infrastructure intervention.

Developer & Cloud Operations | Lumston · Guadalajara

Jul 2022 - Apr 2023

100% remote

Tools and technologies: SQL, AWS Lambda, Amazon S3, FastAPI, Python, Node.js, MongoDB, CSS3, Amazon Connect, contact centers, Contact Lens, and Java Spring with JPA, Hibernate, and JDBC.

- Developed multiple modules for a custom ERP system, including user management, sales, email notifications, and SMS notifications. I used Angular, React, HTML, CSS3, and Tailwind on the frontend, and Python, **FastAPI**, SendGrid, and Twilio on the backend.
- Used NestJS and TypeORM in several projects.
- Developed **serverless microservices** with **AWS Lambda** in Node.js and Python, Amazon API Gateway, Amazon RDS, SNS, DynamoDB, MongoDB, **EventBridge**, and **CloudWatch**.
- Designed and **developed** cloud contact centers and **call centers using Amazon Connect, Lambda Functions**, Amazon Lex, and Amazon Contact Lens. I incorporated artificial intelligence into telephone calls to perform sentiment analysis and improve the user experience.

Senior Full Stack Developer | Clip.mx · Digital payments platform

Dec 2021 - May 2022

100% remote

Tools and technologies: FastAPI, **Python**, Node.js, MongoDB, (HTML / CSS), ReactJs, SQL, **AWS** Lambda, Amazon S3, SNS, Jenkins, CloudFormation, and **Java** with **Spring**, JPA, Hibernate, and JDBC, Jira.

- For the integration of a four-month reporting solution, I used JavaScript, **React**, HTML, and CSS on the frontend, and serverless microservices with **AWS Lambda** in Node.js and Python, API Gateway, SNS, SQS, S3, and **EventBridge** on the **backend**.
- Performed static code analysis with **SonarQube**.
- Implemented **infrastructure as code with AWS** CloudFormation to automate deployments of Lambda functions, SNS, and S3.
- Designed and implemented a **continuous integration and deployment** workflow using Git, GitHub, and Jenkins to deliver services across different environments.
- Enhanced and fixed services previously developed with **Java** and **Spring** Boot, MVC, JPA, Hibernate, and JDBC. In selected cases, I migrated them to **serverless microservices** using **AWS Lambda functions** implemented in Node.js or **Python**.

Full Stack Developer | Innohub Mexico / Rubik Ventures

Apr 2019 - Sep 2021

Portfolio startup collaborations (1-3 months each) · Mexico City

Tools & Tech: Node.js, GraphQL, Socket.IO, **Angular**, **Vue.js,** Next.js, **React**, **Terraform**, **AWS**, Microservices, **Serverless**, SSR.

- **Venture Building Collaboration:** Co-engineered technical foundations for a portfolio of high-impact **startups**, translating the business visions of founders **Katria Oviedo** and **Camilo Manjarr**éz, while operating under the venture ecosystem backed by Gonzalo Hevia Baillères and Raymundo Burguerea.
- **Core Engineering & Architecture**: Part of the core 5-person team that materialized **Numina** from ideation to its eventual acquisition by Santander. Architected custom, scalable solutions integrated with critical financial and regulatory networks (**STP, Cecoban, Banxico, and SAT**).
- **Cloud Infrastructure & DevOps:** Mentored in **advanced AWS architectures by Lead Solutions Architect Miguel Espinosa (Don Jato)**. Implemented Automated CI/CD (Bitbucket Pipelines) and designed **Terraform**-based disaster recovery workflows for **Numina** and **Avanttia**.
- **Real-Time & Frontend Performance:** Engineered low-latency, real-time communication features using GraphQL Subscriptions and WebSockets (**Socket.IO**). Optimized performance via Next.js SSR (**Server Side Rendering**) workflows for modern architecture and integrated **React Helmet** on the **frontend** specifically for dynamic head management in **Client-Side Rendering (CSR)** modules.

Full Stack Developer | Numina · Innohub Mexico / Rubik Ventures

Dec 2019 - Sep 2021

Hybrid · Mexico City

Tools and technologies: SQL, Lambda, SQS, VPC, Amazon S3, FastAPI, Python, Node.js, MongoDB, and CSS3, **GraphQL**, **NextJS, Nest**.

- Managed three websites, two built with **Angular** and one with **React**, implementing JWT security on every request. I automated deployments to Amazon S3 across multiple environments with bitbucket-pipelines.yml, ran unit tests, and applied code inspection with **SonarQube**.
- Developed serverless microservices with **AWS Lambda** and **API Gateway** using Node.js or Python. In API Gateway, I configured an **authorizer** to validate tokens and respond to the appropriate resource and method. I automated deployments, including unit tests and Lambda function code updates through **AWS CLI**.
- **Funds Disbursement System** | **CECOBAN Integration**: Designed a **serverless**, **event-driven workflow** in which uploading a **payroll** or payment **Excel file** to **Amazon S3 triggered an AWS Lambda function through Amazon EventBridge**. The function **read** and **split** the data into batches of up to 10 messages, then sent them to an **Amazon SQS** FIFO **queue** to **preserve processing order** and help prevent duplicate processing. A **second Lambda function processed the messages**, formatted the data to meet CECOBAN's specifications, and sent the resulting flat file over **SFTP from a VPC**, using a **NAT Gateway** and a **static Elastic IP** for outbound connectivity.

Full Stack Developer | Envíoclick · Innohub Mexico / Rubik Ventures

Apr 2019 - Dec 2019

On-site · Mexico City

Tools and technologies: SQL, AWS Lambda, Amazon S3, FastAPI, Python, Node.js, NodeJS, MongoDB, **Looker Studio**, BigQuery,

- Contributed to the migration of an application developed with **Laravel and Vue.js** to a **microservices architecture** using **AWS Lambda** or **Python [V2, V3]**.
- Developed a new landing page with **React**, **Hooks**, **Redux**, Styled Components, and CSS3.
- Fixed defects and added new functionality to an existing Angular application.
- Collaborated with **the UI/UX team to improve existing interfaces**.
- Applied **Clean Code practices** to improve code **readability** and **maintainability**.

Web Developer | Copiloto Satelital · Michelin Group

May 2018 - Apr 2019

On-site · Mexico City

- Designed and developed a **cross-platform Ionic and Angular application** available on Google Play for Android. I was responsible for both the backend and frontend, as well as publishing the application to the store.
- Implemented Ionic, Angular, .NET Web API with REST APIs, Node.js, SQL Server and stored procedures, Git, and Bitbucket.
- **Fixed defects and enhanced a GPS vehicle-monitoring platform using** .NET with Visual Basic and C#, JavaScript, HTML5, CSS3, Web Services, WCF, REST APIs, SQL Server, geolocation through Google Maps API v3, Git, and Team Foundation.
- Contributed to integration services using Web Services and WCF with .NET, ASP, C#, Visual Basic, and SQL Server, as well as microservices with Node.js, MongoDB, Mongoose, and SQL Server.

Web & Desktop Application Developer | Mx-Race

May 2015 - Apr 2018

On-site · Mexico City

- **Designed and developed an electronic timing system using .NET and SQL Server, with the ability to publish live results through Web Services developed in PHP.**
- Designed, **developed**, and implemented a registration system **for athletic races with credit and debit card payment** processing. The system was built with PHP, HTML5, CSS3, JavaScript, MySQL, and .NET, and consumed Web Services.
- During the timing process, I used **RFID chip readers and IPICO Sport technology** for access control and electronic timing. I connected through Telnet with Visual Basic to retrieve RFID chip reads and calculate participant times.

Cisco Field Engineer | Scitum, S.A. de C.V. · Sonda IT

Nov 2014 - Aug 2015

Field work and customer-site visits · Guerrero, Mexico

Performed advanced configuration of Cisco devices, applying CCNA knowledge, routing, VPNs, VLANs, and Call Manager creation for UNINET and Red Uno Telmex.

Field Engineer | Sonda Mexico

Feb 2011 - Dec 2012

Field-based · Guerrero, Mexico

Configured and administered network infrastructure with Cisco routers and switches, VLANs, VPNs, and enterprise connectivity through Red Uno Telmex link management, applying CCNA knowledge.

EDUCATION

B.S. equivalent in Computerized Systems and Information Technology

Universidad Loyola del Pacífico (ULP)

August 2006 - February 2011

Avenida Heroico Colegio Militar s/n, Acapulco, Guerrero, Mexico

Degree pending, Grade average: **9.1**/10

Continuing education

Platzi and Udemy

2014 - 2026

Diplomas and certificates:

University Documents, **Scrum** Developer, **OAWSP Top 10**, AWS **Amazon Web Services**, GCP **Google Cloud Platform**.

https://edgar-colon.mx/education